INCH-POUND
MIL-DTL-45912C
19 May 2021
SUPERSEDING
MIL-DTL-45912B
6 August 2013

DETAIL SPECIFICATION

BOLT, SELF-LOCKING, HEXAGON HEAD, STEEL

Reactivated after 19 May 2021 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers one type of steel, hexagon head, self-locking bolt.

1.2 Classification. The bolt will be furnished in the following compositions:

Composition A – SAE Grade 5
Composition B – SAE Grade 8

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads- UN and UNR Thread Forms
FED-STD-H28/20	-	Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners
MS35764	-	Bolt, Self-Locking, Hexagon Head, Steel, Grade 8, UNC-2A and UNF-2A

(Copies of these documents are available online at https://quicksearch/assist.dla.mil.)

2.3 Non-Government publications.. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from https://www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE-J429	-	Mechanical and Material Requirements for Externally Threaded Fasteners

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 Material. The bolts shall conform to the material composition of SAE Grade 5 or Grade 8 in accordance with SAE J429.

Composition A – SAE Grade 5 (MS35763)
Composition B – SAE Grade 8 (MS35764)

3.2 Design. The bolts shall conform to the design, dimensions and tolerances specified on MS35763 or MS35764 and in Figure 1 and Table I.

3.3 Protective finish. The bolts shall be:

a. Cadmium /1 plated in accordance with Type II, Class 2 of SAE AMS-QQ-P-416.
b. Zinc-Nickel plated in accordance with MIL-PRF-32647 Type K16.
c. Phosphate coated in accordance with MIL-DTL-16232 Type Z, Class 2.

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred alternate.

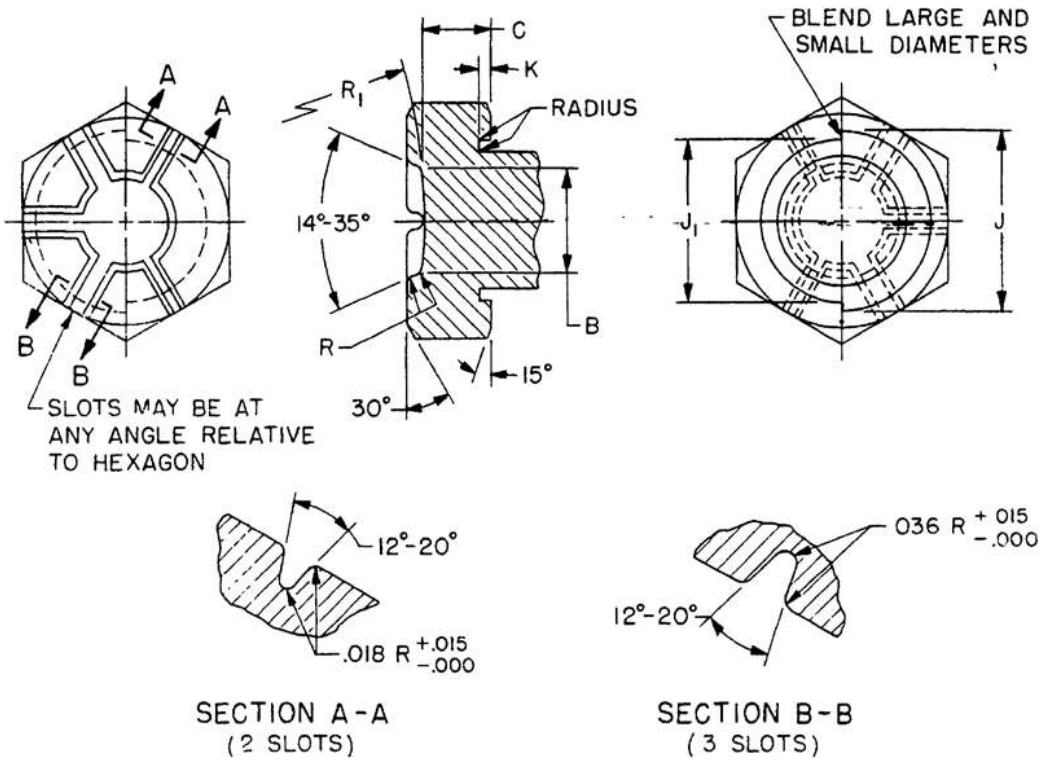


FIGURE 1. Head design.

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TABLE I. Head dimensions.

Nominal Size			1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	7/8	1	1-1/4
B	Recess	Max	.166	.208	.250	.288	.332	.374	.416	.500	.584	.666	.834
	Dia	Min	.146	.188	.230	.268	.312	.354	.396	.480	.564	.646	.814
C	Checking	Max	.110	.145	.185	.205	.220	.260	.290	.335	.390	.455	.590
	Height	Min	.090	.125	.165	.185	.200	.240	.270	.315	.370	.435	.570
J	Recess Dia	Max	.317	.396	.474	.536	.630	.693	.750	.875	1.000	1.125	1.380
	Large	Min	.307	.386	.464	.526	.620	.683	.740	.865	.990	1.115	1.370
K	Recess	Max	.007	.013	.013	.013	.013	.013	.013	.013	.013	.013	.013
	Depth	Min	.004	.010	.010	.010	.010	.010	.010	.010	.010	.010	.010
J_1	Recess Dia	Max	.312	.375	.437	.500	.562	.624	.687	.812	.937	1.062	1.312
	Small	Min	.302	.365	.427	.490	.552	.614	.677	.802	.927	1.052	1.302
K	Recess	Max	.007	.013	.013	.013	.013	.013	.013	.013	.013	.013	.013
	Depth	Min	.003	.006	.006	.006	.006	.006	.006	.006	.006	.006	.006
		Max	.031	.031	.031	.031	.031	.031	.031	.031	.062	.062	.062
R	Radius	Min	.018	.018	.018	.018	.018	.018	.018	.018	.031	.031	.031
		Max	2.00	2.00	2.00	2.00	2.00	2.00	3.50	3.88	4.62	6.88	19.88
R_1	Radius	Min	1.50	1.50	1.50	1.50	1.50	1.50	3.00	3.38	4.12	6.38	19.38

(Wide slots in top of head located with wide recess on underside of head.)
(Narrow slots in top of head located with narrow recess.)

3.3.1 Hydrogen embrittlement. Cadmium plated [1/], Grade 8 bolts specified herein shall be subjected to SAE AMS-QQ-P-416 embrittlement relief requirement as follows:

[1/] Not for new design unless design requirements for the end item exceed the limitations of Zinc-Nickel plating.

Time between plating and baking	Temperature for baking	Time for baking
Shall not exceed 4 hours	375 ±25˚F	23 hours minimum

3.3.1.1 Relief test. In addition, the bolts shall be subjected to the embrittlement relief test of SAE AMS-QQ-P-416.

3.4 Mechanical properties. The mechanical properties of the bolts shall meet the requirements as specified in Table II (see 4.4.3 and 4.4.4).

TABLE II. Mechanical properties.

Composition	Properties	Bolt Size - Inches	
		Up to 1” Inclusive	Over 1”
A	Tensile Strength PSI Min	120,000	105,000
	Proof Load PSI	85,000	74,000
	Hardness Rockwell C Range	25-34	19-30
B	Tensile Strength PSI Min	150,000	150,000
	Proof Load PSI	120,000	120,000
	Hardness Rockwell C Range	32-38	32-38

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3.5 Threads. Threads shall be UNC or UNF, class 2A, as specified on the applicable Military standard in accordance with FED-STD-H28/2. Acceptability of threads shall be in accordance with FED-STD-H28/20, system 22.

3.6 Head.

3.6.1 Head taper. The taper of the head (angle between one side or sides of the head and the axis of the fastener) shall not exceed 2 degrees, with the specified width across the flats being the largest dimensions.

3.6.2 Bearing surface. The bearing surface of the head shall be at right angles to the axis of the body within 2 degrees for bolt sizes up to and including one inch and within 1 degree for bolt sizes larger than one inch. The bearing surface of the head shall be concentric with the axis of the body within 3 percent of the maximum width across the flats. The diameter of bearing surface shall be equal to 95 percent of maximum width across flats within plus or minus 5 percent.

4 VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot size. A lot shall consist of all bolts of the same size, composition, protective finish and thread produced under essentially the same conditions and submitted for inspection at one time.

4.2.2 Sampling for examination. A random sample of bolts shall be taken from each lot in accordance with Inspection Level II of ASQ Z1.4.

4.2.3 Sampling for tests. A random sample of bolts shall be taken from each lot in accordance with Table III. Any bolt which fails to pass any of the tests shall be a defective and cause for rejection of the entire lot represented.

TABLE III. Sample size and acceptance – rejection number for test.

Lot Size	Sample Size	AC	RE
2 – 50	2	0	1
51 – 500	3	0	1
501 – 35000	5	0	1
Over 35000	8	0	1

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4.3 Examination. The sample bolts of 4.2.2 shall be examined as specified in Table IV. Any bolt which contains one or more defects shall be a defective.

TABLE IV. Classification of defects.

Categories	Defects	Inspection Method
Critical	None	
Major		
101	Threads (3.5)	SIE /1
102	Width across flats (3.2)	SIE
103	Undercut depth of head (3.2)	SIE
104	Slot and recess radius of head (3.2)	SIE
105	Bearing surface of head (3.6.2)	SIE
Minor		
201	Overall length (3.2)	SIE
202	Thread length (3.2)	SIE
203	Head height (3.2)	SIE
204	Body diameter (3.2)	SIE
205	Head taper (3.6.1)	SIE
206	Other dimensions (3.2)	SIE
207	Protective finish (3.3)	Visual

/1 SIE = Standard Inspection Equipment

4.4 Tests.

4.4.1 Material. The contractor shall furnish certification the material complies with the requirements specified in 3.1.

4.4.2 Protective finish. The contractor shall furnish certification the protective finish conforms to the requirements of the specifications referenced herein. When required by the procuring agency, the sample bolts shall be tested as specified in the protective finish specifications.

4.4.3 Proof load and tensile strength. The contractor shall furnish certification that the tensile strength and proof load bolts conform to the requirements specified in 3.4. When required by the procuring agency, the sample bolts specified in Table II shall be tested for hardness (see Table I). The Rockwell hardness shall be measured at mid-radius on a transverse section taken one diameter from the point of the bolt.

4.4.4 Hardness. The contractor shall furnish certification the hardness conforms to the requirements specified in Table I. When required by the procuring agency, the sample bolts

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specified in Table II shall be tested for hardness (see Table I). The Rockwell hardness shall be measured at mid-radius on a transverse section taken one diameter from the point of the bolt.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bolt covered by this specification is intended for use when a self-locking fastener is needed or where the reduction of fatigue failure is desired. Typical applications include uses as cylinder head bolts, flywheel housing bolts, connecting rod bolts, and as bearing cap screws. The bolt may be used in tapped holes or with nuts.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Composition (A or B) (see 1.2)
c. Applicable MS part number (see 3.2).
d. Copies and distribution of test certifications.
e. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Grade 5
Grade 8
SAE Grade 5
SAE Grade 8
Zinc-Nickel

6.4 Description. The bolt covered by this specification is a one piece, free-spinning type of self-locking bolt. Locking action is provided by a flexible diaphragm which is formed between the slotted segments of the upper part of the bolt head and the circular recess adjacent to the shank in the lower part of the head. This diaphragm acts as a controlled spring element when the head is wrenched against a rigid seat. When used as a bearing cap screw, the material which the head of the bolt seats should be hard enough to form a rigid seat.

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6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Air Force – 71
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